



SECJ3553 ARTIFICIAL INTELLIGENCE

Section 03

ASSIGNMENT 2 STATE SPACE SEARCH (5%)

Theme: Agriculture

AI-Powered Predictive Analytics for Optimised Crop Yield and Risk Mitigation

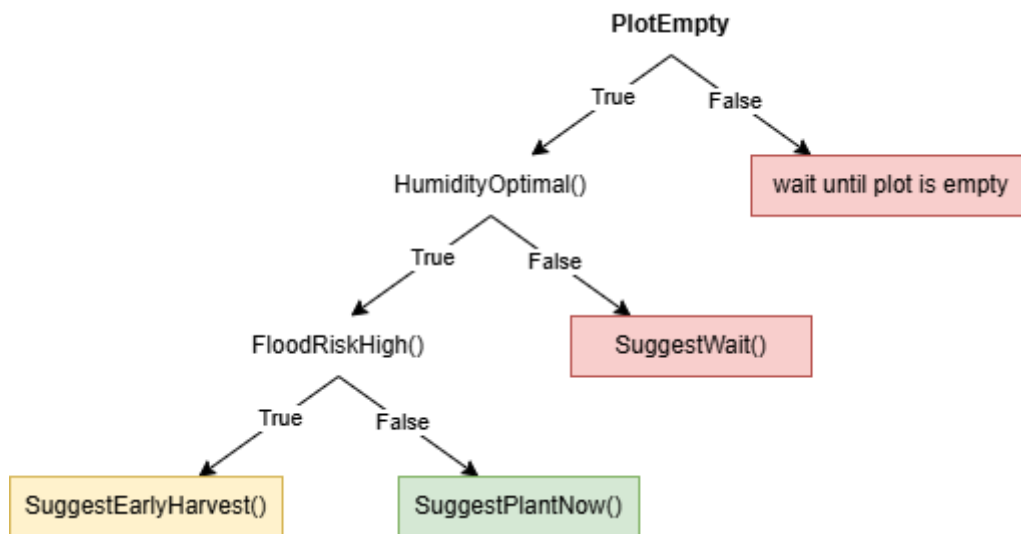
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1.0 PLANTING PHASE OPTIMISATION

1.1 State Space Graph



1.2 Explanation of States

States	Explanation
Plot empty	This is the starting state. It represents the farm before any activity starts. The <i>PlotEmpty()</i> predicate is TRUE. It means the farm land is vacant and ready for new crop cycle.
Humidity low	This state will happen when the sensor detects that the soil or air is very dry. The predicate <i>HumidityOptimal()</i> is FALSE. It is risky to plant seeds because they may become dry.
Rain risk high	This state will happen when the weather forecast predicts heavy rain or floods. The predicate <i>FloodRiskHigh()</i> is TRUE. It is dangerous to grow in this state because the seeds may be washed away.
Conditions ideal	This state means it is the perfect time to plant. This will happen when the plot is empty, the humidity is the best (<i>HumidityOptimal()</i> = TRUE), and the weather is safe (<i>FloodRiskHigh()</i> = FALSE).
Planted	This is the goal state. It represents the successful completion of the planting stage. The system will only reach this state after performing the “plant seeds” action under ideal conditions.

1.3 Explanation of Actions

Actions	Explanation
Wait	This action will be taken when the conditions are not safe. It corresponds to the predicate <i>SuggestWait()</i> . The system will go back to the starting point so that it can be checked again.
Notify and wait	This is a safe action taken when the rain risk is high. The system sends a notification to farmers to warn them not to plant and wait to ensure that they do not waste seeds.
Early Harvest	This emergency action is taken only when <i>HumidityOptimal()</i> = TRUE and <i>FloodRiskHigh()</i> = TRUE. It triggers the predicate <i>SuggestEarlyHarvest()</i> to alert farmers to harvest immediately to prevent total loss.
Plant seeds	This action will only be taken when the system is in the ideal conditions state. It corresponds to the predicate <i>SuggestPlantNow()</i> and moves the system directly to the goal state.

1.4 Problem Formulation and Explanation

Components	Details	Explanation
1. Initial state	Plot empty <i>PlotEmpty()</i> == TRUE	The system start from Day 0 and there was an empty farm plot, <i>PlotEmpty()</i> . The first task of the system is to check the environment before making any decisions. The goal is to transition from this empty state to productive planting state without high risk.
2. Action	Let $S(x)$ = set of action-state pairs $S(PlotEmpty) = \{True: HumidityOptimal(), False: Wait until the plot is empty\}$ $S(HumidityLow) = \{Wait: Plot Empty\}$	The algorithm determines the environmental state by checking the sensor. IF the humidity is low, THEN the system will restrict the planting and force to wait, and back to the beginning state, wait for next day. IF the flood/rain risk is high, THEN the system will trigger a notify action to

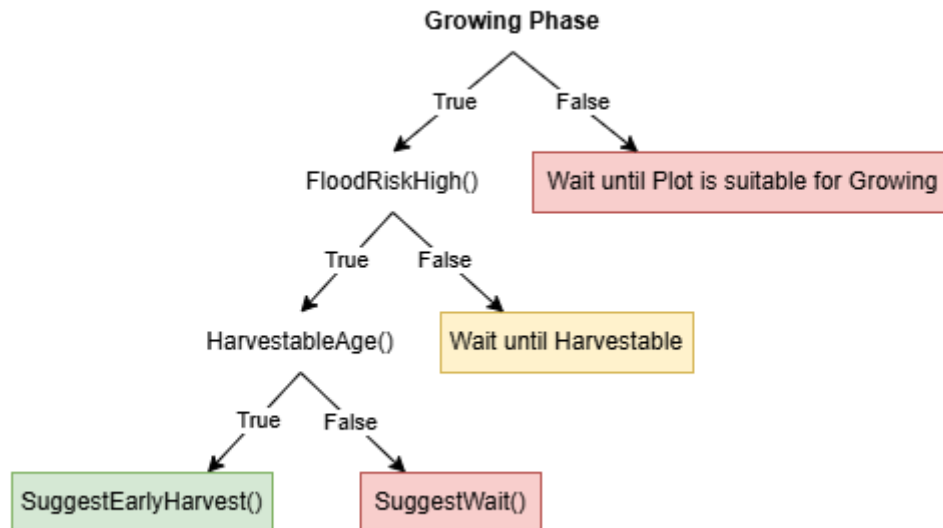
Components	Details	Explanation
	$S(FloodRiskHigh) = \{Wait\ and\ notify: Plot\ Empty\}$ $S(HumidityOptimal() \wedge \neg FloodRiskHigh()) = \{Plant\ seeds: Planted\}$	warn the farmer, and back to the beginning state, wait for next day. IF the conditions are ideals, having optimal humidity and low flood risk, THEN the system allows Plant seed action, and move to the Goal state.
3. Goal test	Planted <i>SuggestPlantNow()</i>	The goal we want to achieve is to successfully plant seeds when the environment is perfect. This ensures that the crop has the best germination rate and is not wasted due to bad weather or dry soil.
4. Path cost	Cost on the route: Time (days taken to reach goal) Each Wait and Notify will add +1 to the cost. Planted costs 0 additional days.	The path cost is the execution time taken to reach goal state. The system tries to minimise this cost by finding the earliest ideal conditions state. If there is delay in planting, it will increase the cost and means that the farmer lost time in the growing season.

Solution:

1. Plot Empty $\rightarrow$ Humidity Optimal $\rightarrow$ Flood/Rain Risk Low $\rightarrow$ Planted
2. Plot Empty $\rightarrow$ Humidity Optimal $\rightarrow$ Flood/Rain Risk High $\rightarrow$ Wait
3. Plot Empty $\rightarrow$ Humidity Low $\rightarrow$ Wait
4. Plot Empty $\rightarrow$ Wait until plot is empty

2.0 EMERGENCY RISK MITIGATION

2.1 State Space Graph



2.2 Explanation of States

States	Explanation
Growing Phase	This is the starting state. The crop is currently in the ground and growing. The system monitors this state daily to protect the crop from sudden environmental threats.
Flood risk low	This state happens when the weather forecast shows safe conditions. The predicate <i>FloodRiskHigh()</i> is FALSE . The crop is safe to continue growing normally.
Risk high but young	This state occurs when there is a flood risk, but the crop is not yet old enough to save (<i>HarvestableAge()</i> = <i>False</i>). Harvesting now would result in zero yield, so the only choice is to notify the farmer and wait.
Risk high & harvestable	This is the critical emergency state. The flood risk is high (<i>FloodRiskHigh()</i> = <i>True</i>) AND the crop is old enough to save (<i>HarvestableAge()</i> = <i>True</i>). This triggers the rescue action.
Crop Saved	This is the goal state. It represents the mitigation of risk. The system reaches this state after performing the "Early Harvest" action, securing the yield before the flood hits.

2.3 Explanation of Actions

Actions	Explanation
Wait	This action is taken when the flood risk is low. It corresponds to the predicate <i>SuggestWait()</i> . The system loops back to the start to monitor the crop again the next day.
Notify and wait	If the flood risk is high but the crop is too young to harvest, the system will notify the farmer of the danger but advise waiting, hoping the crop survives, as harvesting now yields nothing.
Early Harvest	This emergency action is taken only when flood risk is high AND the crop is harvestable. It triggers the predicate <i>SuggestEarlyHarvest()</i> to alert farmers to harvest immediately to prevent total loss.

2.4 Problem Formulation and Explanation

Components	Details	Explanation
1. Initial state	Growing Phase Crop in ground	The system starts with a crop that is currently growing. The goal is to detect risks and transition to a "Saved" state if necessary.
2. Action	Let $S(x)$ = set of action-state pairs. S(Growing Phase) = {True: Flood Risk High(), False: Wait until plot is suitable for growing} S(Flood Risk Low) = {Wait: Growing Phase} S(Risk High Young) = {Notify: Growing Phase}	The algorithm determines the safety state: IF <i>FloodRiskHigh()</i> is False, THEN the system suggests Wait and returns to start. IF <i>FloodRiskHigh()</i> is True but <i>HarvestableAge()</i> is False, THEN the system suggests Notify and Wait . IF <i>FloodRiskHigh()</i> is True AND <i>HarvestableAge()</i> is True, THEN the system allows Early Harvest to move to the Goal state.

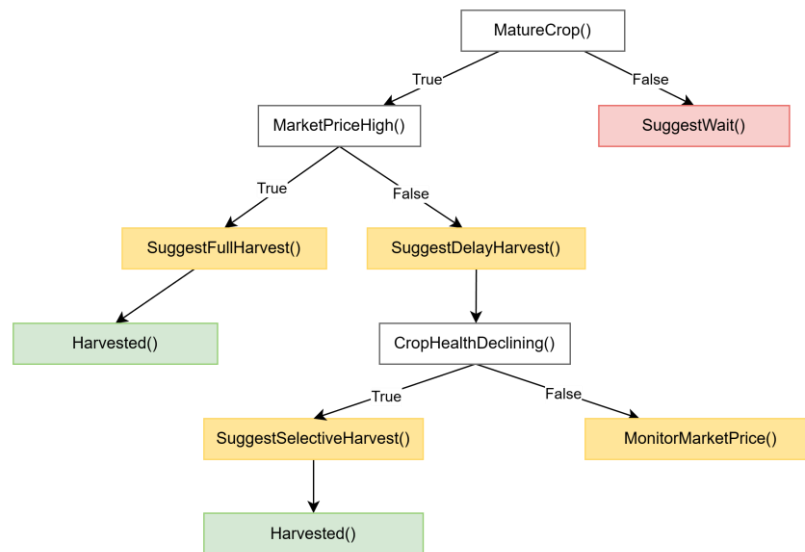
Components	Details	Explanation
	S(Risk High Harvestable) = {Early Harvest: Crop Saved}	
3. Goal test	Crop Saved <i>SuggestEarlyHarvest()</i>	The goal is to successfully harvest the crop <i>before</i> the flood damage occurs. This ensures the farmer gets a yield (even if not optimal) rather than losing everything.
4. Path cost	Cost on the route: Time (days taken) Each Wait adds +1 to the cost. Early Harvest costs 0.	The path cost represents days passed. Minimizing this cost means detecting the risk as early as possible (within the 3-day forecast window) to allow time for harvesting.

Solution:

1. Growing Phase
2. Growing Phase → Flood Risk High → Crop in Harvestable Age → Early Harvest → Crop Saved
3. Growing Phase → Flood Risk Low → Suggest Wait → Flood Risk High → Crop in Harvestable Age → Early Harvest → Crop Saved

3.0 NORMAL HARVEST OPTIMISATION

3.1 State Space Graph



3.2 Explanation of States

States	Explanation
Mature Crop	This is the starting state. This state monitors whether the crops are matured and has reached harvestable age.
Not Mature	The crop is still growing so the system suggests waiting and harvesting now is not recommended.
Market Price High	This state tells if the current market price for the crops planted.
Suggest Full Harvest	This state occurs when the market price is high and the crop is mature. This condition will yield the most profitable harvest, so it is recommended to do a full harvest.
Suggest Delay Harvest	This state suggests for the harvest to be delayed, as it is not the most profitable and optimal harvest time.
Crop Health Declining	This state is to keep track of crop health after delaying harvest.
Suggest Selective Harvest	This state suggests for a selective harvest when CropHealthDeclining() is TRUE so that there can still be profits before the crops are spoiled.
Monitor Market Price	This state is to continue monitor for the best market price for the most profitable harvest.

3.3 Explanation of Actions

Actions	Explanation
Wait	Action taken when crop is not matured yet. It corresponds to the predicate <i>SuggestWait()</i> . The system waits until maturity before further decision.
Full Harvest	Executed when the crop is mature and market price is high. Leads directly to the goal state.
Delay Harvest	Action taken when the crop is mature, but market price is low. System delays harvest and continues monitoring.
Selective Harvest	Executed if delaying causes health risks. A partial harvest is done to prevent loss.
Continue Monitor Market Price	Action that continues observing the price while harvest is delayed and health is still stable.

3.4 Problem Formulation and Explanation

Components	Details	Explanation
1. Initial state	Mature Crop Crop is fully grown	The system starts by checking if the crop is matured and ready to be harvested.
2. Action	Let $S(x)$ = set of action-state pairs. S(Mature Crop) = {True: MarketPriceHigh, False: SuggestWait} S(Market Price High) = {True: Full Harvest, False: Delay Harvest} S(Crop Health Declining) = {True: Selective Harvest, False: Monitor Market Price}	IF <i>MatureCrop()</i> is False, THEN the system suggests Wait and returns to start. IF <i>MatureCrop()</i> is True but <i>MarketPriceHigh()</i> is True, THEN the system allows to do a Full Harvest and move to goal state. IF <i>MatureCrop()</i> is True but <i>MarketPriceHigh()</i> is False, THEN the system will delay the harvest. IF <i>CropHealthDeclining()</i> is True, THEN the system allows for Selective Harvest and move to the Goal state.

Components	Details	Explanation
		IF <i>CropHealthDeclining()</i> is False, THEN the system allows will continue monitoring for the best market price.
3. Goal test	Harvested	The goal is to get the most optimal and profitable harvest by harvesting at the best possible timing and conditions.
4. Path cost	Cost is increased each time the harvest is delayed.	The path cost represents the loss in profit and increased risk caused by delaying harvest. Each time the system chooses to wait, crop quality may decline, and market prices may change, leading to higher overall cost. Therefore, the optimal path is the one that reaches the harvest state with the least delay.

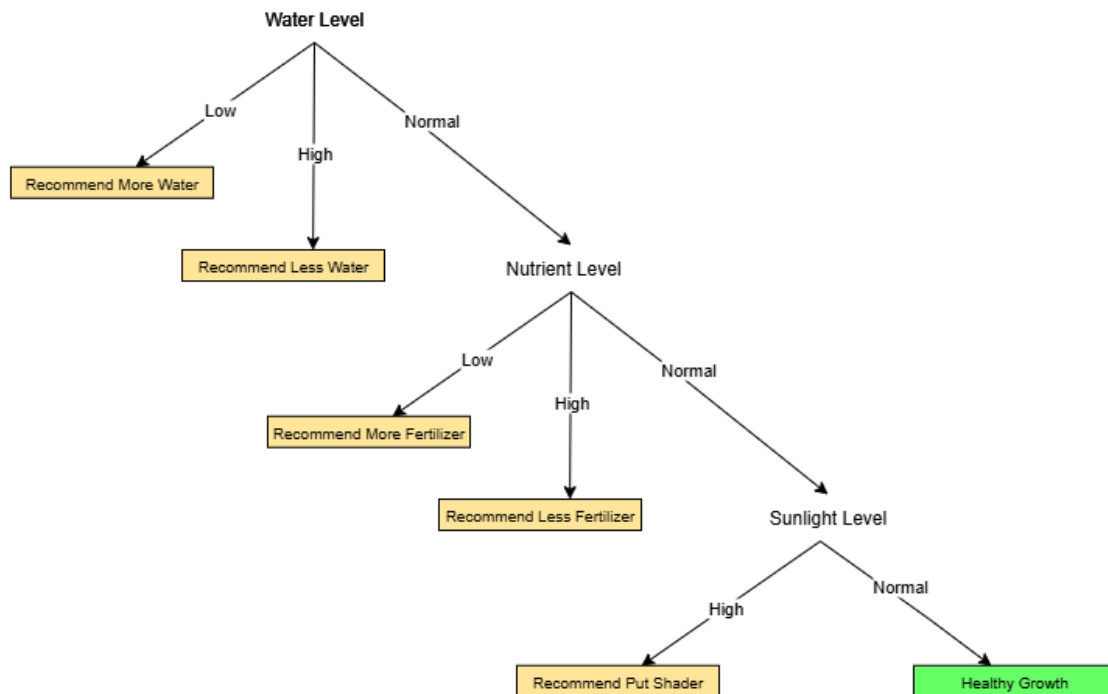
Solution:

Mature Crop → High Market Price → Full Harvest → Harvested

Mature Crop → Low Market Price → Delay Harvest → Crop Health Decline → Selective Harvest → Harvested

4.0 RESOURCE & GROWTH MONITORING

4.1 State Space Graph



4.2 Explanation of States

States	Explanation
Water Level	This is the starting state. This state is to monitor the water level of the crop daily.
Recommend More Water	This state happens when the water level detected is low, it might suggest adding more water on the next day.
Recommend Less Water	This state happens when the water level detected is high, it might suggest lesser the water given on the next day.
Nutrient Level	After checking the water level to be normal, this state will start to monitor the nutrient level of the crop.

States	Explanation
Recommend More Fertilizer	This state happens when the nutrient of the crop detected to be low, it will recommend the farmer to add more fertilizer to the crop on the next day.
Recommend Less Fertilizer	This state happens when the nutrient of the crop detected to be high, it will recommend the farmer to give less fertilizer to the crop on the next day.
Sunlight Level	This state is used to check that the sunlight level would be over high or not.
Recommend Put Shader	This state happens when the sunlight is high, it will recommend putting a shader above the crops
Healthy Growth	This is the goal state. This state shows that the crops is healthy growing as the conditions are all showing normal.

4.3 Explanation of Actions

Actions	Explanation
Low	This action is for state checking condition to be low condition.
High	This action is for state checking condition to be high condition.
Normal	This action is for state checking condition to be normal condition.

4.4 Problem Formulation and Explanation

Components	Details	Explanation
1. Initial state	Water Level	This module will start with monitoring the water level of the crops. By the result given, it will recommend the best way to maintain the crops for healthy growth.

Components	Details	Explanation
2. Action	Let $S(x)$ = Set of action-state pairs. S (Water Level) = {Low: Recommend more water, High: Recommend less water, Normal: Nutrient Level} S (Nutrient Level) = {Low: Recommend more fertilizer, High: Recommend less fertilizer, Normal: Sunlight Level} S (Sunlight Level) = {High: Recommend put shader, Normal: Healthy growth}	The system helps user to monitor the growth of the crops. The system will start with the water level monitoring first. If the water level is low, system will recommend adding more water on the next day, but if the water level is high, system will recommend give lesser water next day. When the water level shows normal, it will move to the next monitoring cycle. Next is the nutrient level monitoring, same as the water level, when low nutrient, add more fertilizer next day. When crops have high nutrient, give less fertilizer. When the crops show normal nutrient level, move to the last monitoring cycle, which is sunlight level. When system detect sunlight level is too high, system will recommend putting a shader above the crops. When it shows normal, means the crops is healthy growth.
3. Goal test	Healthy Growth	The goal here is to make sure the maximum rate of healthy growth crops. This will help farmer to improve the efficiency as using technology to monitor crops can make decision correctly.
4. Path cost	Number of monitoring steps taken before reaching Healthy Growth	Each state change contributes one unit of cost. If the condition is abnormal (low or high), the system makes a recommendation and waits for the next monitoring cycle, resulting in additional cost.

Components	Details	Explanation
		The optimal path is the one where all conditions are normal, thus reaching Healthy Growth with the minimum cost.

Solution: Water Level → Nutrient Level → Sunlight Level → Healthy Growth